QuantumLoop ERP - Foundation Document v1

Oct 20, 2026 · @QuantumLoop

1. How to use this document

This is the single reference for what we build, how we build it and how we sell it. It replaces the earlier, longer notes.

Read sections 2 to 5 for the product, sections 6 to 9 for the engineering approach, and sections 10 to 12 for pricing, migration and sequence. Section 13 lists the decisions the team must settle, and by when.

Anything marked as a hypothesis (prices, timelines) is a starting point to test, not a commitment.

2. Product vision and scope

We are building one platform for Indian small businesses that starts with sales and extends to accounts and HR, sold as multi-tenant SaaS.

Who it is for. Indian businesses with 5 to 50 employees that run a telecalling team, a field sales team or both, and earn repeat revenue from renewals such as AMC, subscriptions and service contracts. QuantumLoop is customer zero: we run our own sales on it first. The first external customers are QuantumLoop's existing clients.

Working name. The current screens are branded Lead Desk. This document uses "QuantumLoop ERP" as a placeholder; section 13 settles the final name.

What v1 includes

- Leads, companies and contacts, with list, kanban and activity history
- Telecaller and field-sales capture with manager approval
- Contracts, renewals, invoices and payments (sales accounting)
- Employee records, attendance and leave
- Tasks, projects, dashboards and audit logs

What v1 deliberately excludes

- Statutory accounting: general ledger, GST return filing, e-invoicing, e-way bills. We integrate with Zoho Books or Tally instead.
- Payroll with PF, ESI, TDS and professional tax
- Inventory, purchasing and manufacturing
- Marketing automation and email campaigns

The excluded items carry heavy compliance load, change with every budget, and already have mature products. Customers want them connected, not replaced.

Design principles

1. The company record is the hub: leads, contracts, invoices and visits all hang off it.
2. Where the business needs control, nothing becomes official until a manager approves it.
3. Every record has a tenant, an owner and an audit trail.
4. Finish one module before starting the next.

3. User roles and permissions

Nine tenant roles cover every user we expect in a small business, plus one platform role for QuantumLoop staff.

Role	Typical person	Can do	Cannot do
Owner	Business owner	Everything in the tenant, including billing and plan	Nothing is blocked
Admin	Office or IT lead	Users, roles, settings, import and export, all modules	Change billing or delete the tenant
Manager	Sales or branch manager	See team data, approve visits and telecaller leads, assign leads, run reports	Manage users or settings
Sales Executive	Inside sales	Own leads, companies, tasks and draft invoices	See others' leads unless shared, approve anything
Telecaller	Calling staff	Log calls, submit leads for approval, see own follow-ups	Edit approved leads, see other callers' leads, export data

Role	Typical person	Can do	Cannot do
Field Executive	Visiting staff	Start and end sessions, record visits with photo and location, see own follow-ups	Edit a submitted visit, see others' visits
Accountant	Accounts staff	Contracts, renewals, invoices, payments, financial reports	Change the lead pipeline, see HR records
HR	HR staff	Employee records, attendance, leave, documents	See invoices or lead data
Viewer	Auditor, partner	Read-only on assigned modules	Any change
Platform Admin	QuantumLoop support	Create tenants, set plans, run support sessions	Open tenant data outside a logged support session

How permissions work

- Permissions are small named actions such as `leads.approve`, `visits.approve` and `invoices.create`. A role is a bundle of permissions.
- Each permission carries a data scope: own records, team records or all records.
- Phase 1 ships these fixed roles. Custom roles become a paid-plan feature later.
- One user holds one role per tenant in v1.
- A job title such as "Customer Relations Manager" is only a label. The role decides access.
- Every approval, export, role change and support session is written to the audit log.

4. Modules and features

Seven modules make up the platform. Three are running today in some form; telecaller approval, contracts and renewals, and HRMS are new.

Module	Exists today	New work	Phase
CRM: leads, kanban, companies	List and kanban views, companies, overdue and today's follow-ups	Duplicate detection, bulk actions, custom fields	1
Field Sales	Sessions with live location, visit photo, outcomes, manager review queue	Offline mode, bulk approve, location checks	1
Telecaller	Performance table on the dashboard	Call log, lead submission, manager approval	1
Accounting and renewals	Invoices, invoice settings	Product catalogue, contracts, renewal tracker, payments, ageing	2
Projects and tasks	Projects, My Tasks	Link tasks to company and contract	2
Platform and admin	Users, activity logs	Tenants, plans, billing, import wizard, audit log	0 and 3
HRMS	Nothing	Employee master, attendance, leave, documents	4

Telecaller (new)

- Call log with lead, outcome, duration, notes and next follow-up date.
- "Add lead from call" creates the lead as Pending approval.
- A duplicate check on the normalised phone number runs before submission.
- A manager approves or rejects with a reason. Only approved leads count in the pipeline and in telecaller metrics.
- Daily call and lead targets per telecaller appear on the dashboard.
- Later: click-to-call and automatic call logging through cloud telephony.

Field Sales (harden what exists)

- The dashboard shows 58 visits awaiting review. Add bulk approve, filters by executive and date, and a review-time target.
- Visits save on the phone and sync when the network returns.

- Photos are compressed on the device. Each visit records location accuracy and flags mock-location apps.
- Sessions close automatically at day end, with the end source recorded.
- The visit outcome sets the next follow-up date.

Accounting and renewals (new)

This module answers two questions: which clients need to renew, and what each client has bought.

- Product and service catalogue with HSN or SAC code, GST rate and default price.
- One contract per purchase: company, product, start date, end date, billing cycle, amount, auto-renew flag.
- Renewal tracker for contracts expiring in 30, 60 and 90 days, with revenue at risk and an owner for each renewal.
- Reminders to the owner and the client contact before expiry.
- Invoices linked to contracts, with GSTIN, place of supply and the CGST, SGST or IGST split.
- Payments by UPI, bank transfer, cash or cheque, including part payments, with an outstanding and ageing report.
- Export to Zoho Books and Tally formats. No ledger and no GST filing in v1.

HRMS (new)

- Employee master: profile, documents, emergency contact, bank details.
- Organisation structure: department, designation, reporting manager.
- Attendance by check-in and check-out. Field Sales sessions post attendance automatically.
- Leave types, balances, requests and approval, plus a holiday calendar per tenant.
- Onboarding and exit checklists, with offer and appointment letter templates.
- An employee record is separate from a login, so staff without computer access can still be recorded.
- Payroll, PF, ESI and TDS are out of scope. Attendance exports to the customer's payroll tool.

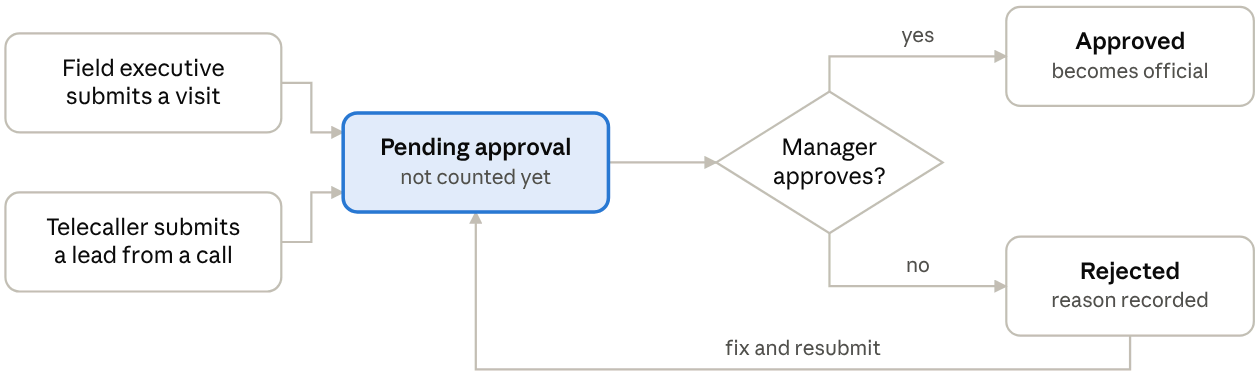
Platform and admin

- Tenants, plans, usage limits and billing.
- Import wizard (section 11), audit log and, later, custom roles and API keys.

5. Core workflows

Field visits and telecaller leads share one approval flow, leads follow the stages already on the dashboard, and renewals run on a countdown.

One manager approval serves field visits and telecaller leads



approval flow · 2 entry points, 1 decision

A rejected item returns to its submitter with a reason. An approved item becomes official and counts in reports and metrics.

Lead lifecycle. Stages stay as they are today: New, Attempted, Contacted, Follow Up, Interested, Qualified, Proposal Sent, Negotiation, Won and Lost. Every stage change is written to history. A lead can move to Won only with a company attached, and Won opens a contract when a product was sold.

Approval rules

- A submitter can edit a record only while it is pending or after a rejection.
- A manager cannot approve their own submission. An Admin can.
- A rejection needs a reason, and the submitter is notified.
- A pending item older than two working days is flagged to the manager (hypothesis).
- Approved records are locked. A change needs a new request or an Admin edit that is written to the audit log.

Renewal tracking

1. A contract is created when a lead is won, or entered directly.
2. The tracker lists contracts expiring within 90, 60 and 30 days.
3. Reminders go to the owner at 90, 60, 30 and 7 days, and to the client contact at 30 and 7 days.

4. The owner marks the renewal won, with a new term and invoice, or lapsed, with a reason.
5. Lapsed contracts stay visible for win-back follow-up.

6. Tech stack and engineering standards

Keep Flask and restructure it; the gaps are multi-tenancy, tests and operations, not the framework.

Layer	Recommendation	Why
Backend	Flask as a modular monolith: one package per module, SQLAlchemy 2, Alembic migrations	The team already knows it. One deployable unit is right for three engineers.
Database	PostgreSQL	Row-level security for tenant isolation, JSONB for custom fields. Confirm what the app uses today.
Background jobs	Redis with a job queue (RQ or Celery)	Renewal reminders, imports, emails and exports must not block web requests.
Frontend	Server-rendered templates with HTMX and Alpine.js	Fast to build, no separate single-page app to maintain.
Field mobile	Progressive web app: installable, camera, geolocation, offline queue	No app-store gate and works on any phone. Go native only if the PWA hits a hard limit.
File storage	S3-compatible object storage in an India region, private buckets, signed URLs	Visit photos and employee documents.
Login	Email and password with OTP, Google sign-in as an option, TOTP for owners and admins	Login is Google-only today, and not every customer uses Google accounts.
Email	AWS SES or Zoho ZeptoMail for transactional mail	Invites, renewal reminders, approval notices.
Our own billing	Razorpay subscriptions with UPI AutoPay	Matches how Indian SMBs pay.

Layer	Recommendation	Why
Hosting	Containers in an India region (Mumbai or Hyderabad)	Latency and the data-residency promise in section 9.
Monitoring	Sentry for errors, uptime checks, structured logs, database metrics	We should know before the customer does.
Addresses	Store latitude and longitude; resolve an address once and cache it	Per-call geocoding fees add up at field-sales volumes.

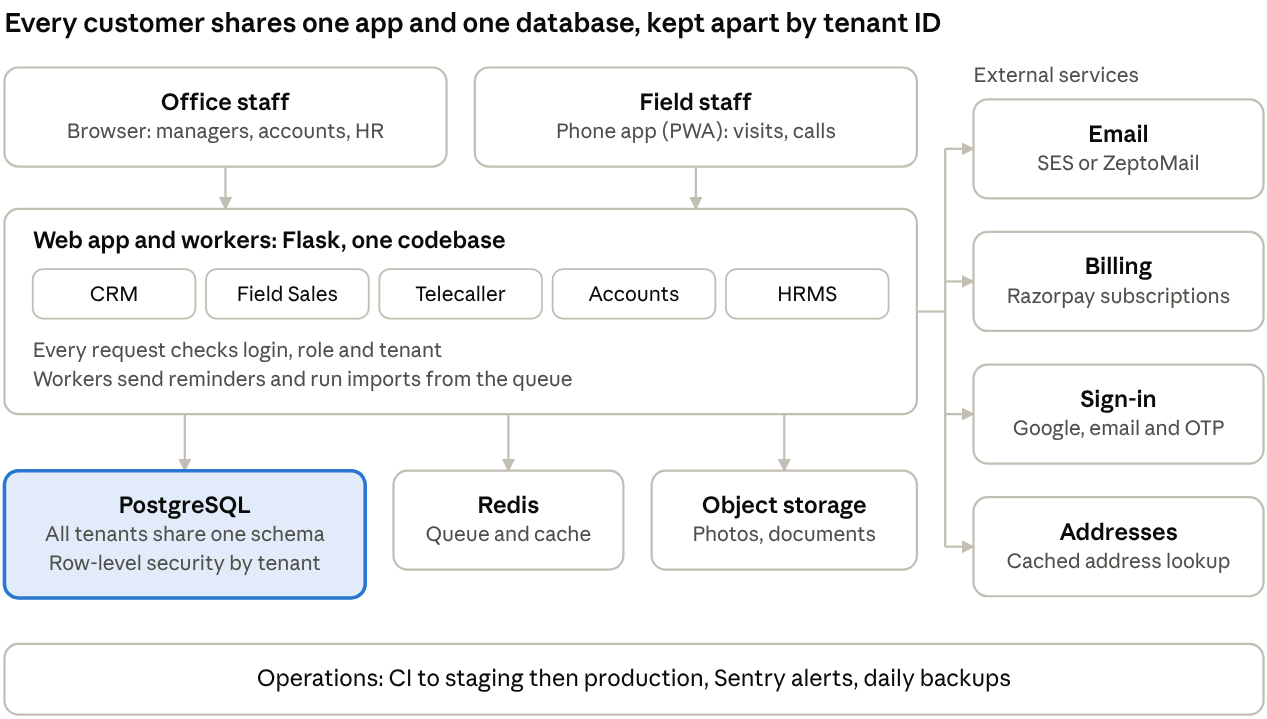
Flask or rewrite? Keep Flask unless a two-day spike in Phase 0 shows the refactor costs more than a rewrite. A rewrite of working modules costs months and adds no customer value. Treat this as a hypothesis to test, and decide in section 13.

Engineering standards: the definition of done for every change

- No secrets or configuration in the repository. Use environment variables or a secrets manager, and rotate any secret that was ever committed.
- Every change goes through a pull request with one reviewer. Pushes to main are blocked.
- Business rules (approvals, renewals, permissions) have automated tests. The main user flows have end-to-end tests. CI blocks the merge when a test fails.
- Every endpoint has a test proving a user from tenant A cannot read tenant B.
- Database changes ship only as migrations, each with a restore plan.
- Deploys run from CI to staging first, then production, with a health check and automatic rollback. Today a push to main deploys straight to production; staging and branch protection are Phase 0 work.
- Dependencies are pinned and scanned. No deprecated libraries.
- Logging, error tracking and uptime alerts exist before any customer is onboarded.
- Each module has a short README and a permission list. Deploy, restore and incident runbooks live in the repository.

7. SaaS architecture

Every customer shares one app and one database, kept apart by a tenant ID and row-level security.



architecture · 1 app, 3 data stores, 4 external services

The database rules, not application code alone, stop one customer from reading another's rows.

Tenancy decision. One shared database and one shared schema is the cheapest to run and the simplest to migrate for dozens of small tenants. Schema-per-tenant or database-per-tenant costs more and complicates every migration, so we reserve it for a customer who pays for isolation. The tenant is resolved from the signed-in user and set on every database session, so the row-level rules apply to every query.

Environments. Local, staging and production. Staging holds a sandbox tenant with synthetic data and never real customer data.

Deployment. CI builds a container image, deploys it to staging, runs smoke tests, then promotes to production behind a health check. Rolling back means redeploying the previous image.

Backups. Daily automated database snapshots with point-in-time recovery, versioned object storage, and a restore test every month. Targets to confirm: lose at most 15 minutes of data and restore within 4 hours.

Limits per plan. User count, storage and API rate limits are enforced per tenant from the plan record.

Observability. Error tracking, uptime checks, structured logs carrying the tenant ID, a slow-query log, and an alert on queue length.

Scaling path. One app server and one database should carry the first tenants. Add app instances and a read replica as load grows, and split the workers onto their own servers first.

8. Data model overview

The company is the hub entity, and one generic approval record serves both field visits and telecaller leads, so the approval logic is built once.

Entity	Key fields	Connects to
Tenant	name, plan, timezone, status	Owns every other record
User	email, role, status	Tenant, Employee (optional)
Employee	name, department, designation, manager, join date, documents	Tenant, User
Company	name, GSTIN, address, phone, owner, source	Contacts, Leads, Contracts, Invoices, Visits
Contact	name, phones, email, designation	Company
Lead	status, owner, source, next follow-up	Company, Activities, Approval
Activity	type (call, note, task), outcome, notes, due date	Lead, Company
CallLog	caller, outcome, duration	Lead, Approval
FieldSession	executive, start and end time and location, end source	Visits
Visit	photo, latitude, longitude, accuracy, outcome, captured time	FieldSession, Company, Approval
Approval	type, record, submitter, reviewer, status, reason, decided time	Visit, Lead

Entity	Key fields	Connects to
Product	name, HSN or SAC, GST rate, default price	Contract
Contract	start, end, billing cycle, amount, auto-renew flag	Company, Product, Invoice
Invoice and InvoiceLine	number, GST split, status	Contract, Payment
Payment	amount, mode, reference, date	Invoice
Attendance and LeaveRequest	date, status, leave type	Employee
AuditLog	actor, action, record, before, after, time	Every entity
ImportJob and ImportRow	source, mapping, status, raw row	Any imported entity

Rules every table follows

- Each row carries `tenant_id`, `created_by`, `created_at`, `updated_at` and a soft-delete flag.
- Money is stored in paise as whole numbers, never as decimals or floats.
- Timestamps are stored in UTC and shown in the tenant's timezone, which defaults to Asia/Kolkata.
- Phone numbers are stored as typed and in a normalised international form, for matching and duplicate checks.
- A GSTIN is validated by format and checksum when saved.
- Imported rows keep `source_system` and `external_id`, so a re-import updates records instead of duplicating them.
- Status changes, such as lead stage and approval decisions, are written as history rows and never overwritten.

9. Security, privacy and Indian compliance

Our customers decide why they collect lead and employee data, so they are the data fiduciaries and we are their processor. That makes security and privacy features part of what we sell.

India's DPDP Rules were notified on 13 November 2025, and full obligations apply from 13 May 2027, per summaries from Bar & Bench and Uniqus. The Act puts duties on the fiduciary and expects them to bind processors by contract, as Hogan Lovells explains. Expect customers to ask us for a data processing agreement. Have a lawyer and a CA confirm the points below before we publish terms.

Security requirements

- Tenant isolation through `tenant_id` on every row, Postgres row-level security and automated cross-tenant tests.
- Encryption in transit and at rest. Bank details, ID numbers and salary are encrypted per field and visible only to HR and the Owner.
- Passwords hashed with Argon2 or bcrypt, OTP and TOTP, session timeouts, rate limits on login and OTP.
- An append-only audit log for approvals, exports, role changes and support access.
- Photos and documents in private buckets, served by signed links that expire in minutes.
- Encrypted backups with a restore test every month.
- Support staff enter a tenant only through a time-limited, logged session that the tenant admin can see.
- Dependency scanning on every build, and a third-party penetration test before we sell widely.

Privacy requirements (DPDP)

- A standard data processing agreement ready for every customer.
- Per-tenant export of all data, deletion on request, and configurable retention.
- Telecallers and field staff capture personal data of third parties. Record the source and a consent note on each lead, and support a do-not-contact flag.
- Field Sales tracks employee location. Track only during an active session, tell staff in the app, and never track outside a session.
- Keep data in an India region. Publish the list of sub-processors: hosting, email, error tracking.
- A breach runbook that works to a 72-hour clock.

Indian business compliance

- Invoices carry supplier and buyer GSTIN, place of supply, HSN or SAC code and the CGST, SGST or IGST split.

- Invoice numbers run in a series that is unique per financial year (April to March), and reports use the Indian financial year.
- E-invoicing and e-way bills are out of v1. Customers who need them keep issuing through their accounting tool, and we add an integration with an approved GST provider later.

10. Pricing and packaging for Indian SMBs

Price per organisation with users included, starting at ₹2,499 a month, so a five-person team pays about ₹2,500 against ₹4,000 for Zoho CRM Standard alone. All figures below are hypotheses to test with five existing clients before we publish anything.

What buyers compare us against (₹, excluding 18% GST, annual billing; checked 20 October 2026)

Product	Covers	Price	Source
Bigin by Zoho	Small-business CRM	Free; ₹400, ₹720 or ₹1,080 per user per month	Bigin
Zoho CRM	Full CRM	₹800 Standard, ₹1,400 Professional per user per month	Aaxonix, citing Zoho's March 2026 price sheet
Freshsales	CRM	₹749 Growth, ₹3,239 Pro per user per month; other sites quote higher	itforsme
LeadSquared	Call-heavy and field sales	Quoted per user on annual contracts	itforsme
Odoo	ERP modules	₹725 Standard and ₹1,350 Custom per user per month at list price	OEC.sh
Zoho Books	Accounting with GST	₹749 to ₹8,332 per organisation per month on paid plans, 3 to 15 users included	Patron Accounting

Product	Covers	Price	Source
Zoho People	HRMS	About ₹50 to ₹230 per employee per month; sources vary and payroll is extra	itforsme

A five-person team on Zoho CRM Professional plus Zoho Books Professional pays about ₹8,250 a month before HR, GST and setup (5 × ₹1,400 plus ₹1,249). That stack has no field-visit approval and no telecaller approval flow.

Why price per organisation, not only per user. Telecallers and field executives are many and low-value seats, so per-user pricing punishes exactly the customers with big calling teams. Our costs grow with data and photos, not with logins. One price for the team is also easier to sell to an owner.

Proposed plans (₹ per month, billed annually, excluding GST)

Plan	Price	Users included	Modules	Storage
Starter	2,499	5	CRM, Telecaller, Tasks	10 GB
Growth	5,999	15	Starter plus Field Sales, Contracts and renewals, Invoices	50 GB
Business	11,999	40	Growth plus HRMS, custom roles, API, priority support	200 GB

- Extra users cost ₹249 per user per month. A cheaper light-user seat for telecallers and field staff is an option later; Odoo already sells one at ₹195.
- Monthly billing costs 20% more than annual.
- Photo storage is the main variable cost. Five field executives making 20 visits a day produce roughly 8 GB of compressed photos a year (assumption: 0.3 MB a photo). Photos older than 24 months move to cheaper archive storage.

One-time onboarding and migration fee. ₹5,000 for Starter with template import, ₹15,000 for Growth, ₹30,000 or more for Business or messy data, quoted after a data check. Zoho partners quote ₹15,000 to ₹50,000 to set up Zoho Books, per Patron Accounting, so this is in range.

Commercial rules

- Prices exclude 18% GST. Business customers claim it back as input credit.
- Annual prepayment is the default. Quarterly through UPI AutoPay (Razorpay) is the fallback.
- No permanent free plan, because support is our real cost. Existing clients get a 30-day pilot.
- Founding offer: the first 10 customers get 25% off for two years in return for feedback and a case study.
- Add the subscription to the annual AMC invoice that QuantumLoop already sends existing clients.

Does it fund itself? At 30 customers (10 Starter, 15 Growth, 5 Business) revenue is about ₹1.75 lakh a month before GST. That should cover infrastructure and one support person. Check the infrastructure figure against real usage after Phase 3.

How we win. We cannot beat Bigin's ₹400 entry price on price alone. We win on field-visit and telecaller approval, renewal tracking, local support and WhatsApp-first onboarding.

11. Data migration

Build one import wizard that every customer uses, add a source template for each tool our customers already run, and sell migration as a paid service.

Migration A: QuantumLoop's own data. Move the current database into the multi-tenant schema as tenant 1. Add the tenant row, backfill `tenant_id`, compare row counts and totals in staging, and keep the old database read-only for 30 days. This is the exit test for Phase 0.

Migration B: customers' data from other tools

Source	What comes across	Method	Priority
Excel or Google Sheets	Leads, companies, contacts, products, employees	Our CSV and XLSX templates	1
Zoho CRM and Bigin	Leads, contacts, accounts, notes	CSV export, mapped	2
Tally	Customers, items, open invoices	Excel or XML export. No ledgers.	2

Source	What comes across	Method	Priority
Freshsales, LeadSquared, HubSpot	Leads, contacts, activities	CSV export	3
IndiaMART and JustDial lead exports	Leads	CSV, with an API pull later	3
Phone and Google contacts	Contacts	CSV or vCard	3
Zoho People, greytHR, Keka	Employee master, leave balances	CSV export, built with HRMS	4

How the import wizard works

1. Upload a CSV or XLSX file and pick a source template, or start blank.
2. Map columns to fields. The mapping is saved for reuse.
3. Validate: normalise phone numbers, check GSTIN, email and dates, and flag duplicates against existing records and within the file.
4. Dry run: show new, updated, skipped and failed row counts, with a downloadable error file.
5. Commit as one import batch. Every row keeps its source name, original ID and raw values.
6. Roll back a whole batch within 14 days if needed (hypothesis).

Data rules

- Match duplicates on normalised phone, email, GSTIN and a fuzzy company name. Suggest merges and never merge automatically.
- Map old pipeline stages to new ones through a table the customer approves.
- Keep original created dates and owners where the owner exists. Otherwise assign the record to an Admin.
- Notes and call history import as activities. Attachments import only on request.
- Rows that fail validation go to a needs-fixing list and are never dropped silently.

Why validation is not optional. Our own data already shows it. The dashboard lists a lead with the phone number "1" and several leads named only "Office" or "Reception" with landline numbers. It also mixes Indian and Australian numbers, so normalisation must handle +91, +61 and STD-code landlines such as 040.

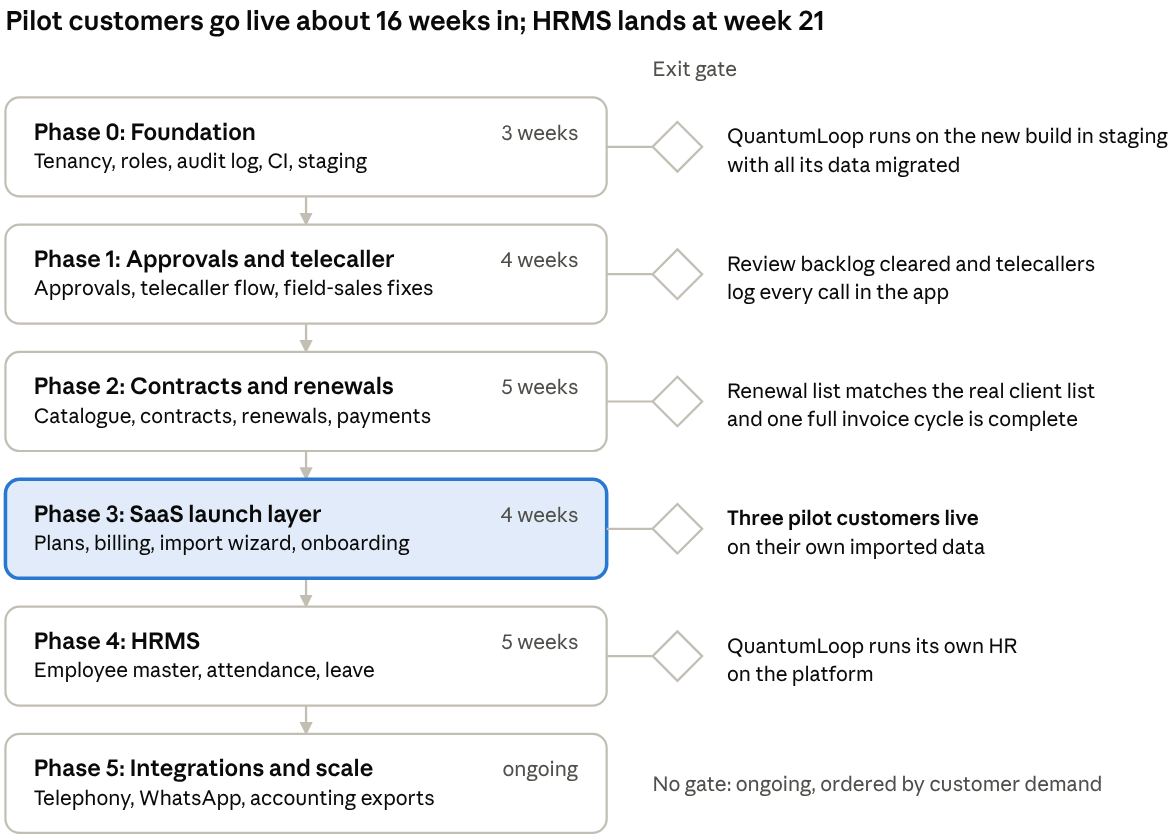
Runbook for each customer

1. Discovery, about half a day: which tools, how many records, who owns the data, what must carry over.
2. Export from the old tool and keep an untouched copy.
3. Clean and map in a sandbox tenant.
4. Trial import. The customer checks 50 sample records against the old system.
5. Cutover: freeze edits in the old tool, import changes made since the trial, switch users over.
6. Run both for one to two weeks and compare record counts and renewal dates.
7. Retire the old tool and keep its export for 90 days.

Done means: record counts match the source apart from rows in the error file, 20 renewal dates spot-checked, and an automated check shows no record landed in the wrong tenant.

12. Phased roadmap

Pilot customers can go live about 16 weeks in, and HRMS follows at about week 21.



roadmap · 6 phases, 5 gates

Each gate is a test we can run, such as migrating QuantumLoop's data or signing three pilot customers.

These estimates assume three engineers working full time and are accurate to about 50% either way (hypothesis). Re-plan at the end of Phase 0, once the Flask spike from section 6 is done.

Rules for the sequence

- Phase 0 is not optional. Tenancy, roles and audit logging are far harder to add after customer data exists.
- A phase ends when its gate is met, not when its date arrives.
- HRMS follows the SaaS launch layer so we can start selling before the largest new module is built.
- Phase 5 items are pulled in one at a time by customer demand.

13. Open decisions

Eight decisions need a final answer, each by the point in the last column. Each has a recommendation; write the final choice and its date beside it once the team agrees.

Decision	Options	Recommendation	Decide by
Product name	QuantumLoop ERP, QuantumLoop Business Suite, keep Lead Desk	Business Suite as the name, with "ERP" in the description. ERP promises inventory, payroll and GST filing that v1 excludes.	Before Phase 3
Flask or rewrite	Keep and restructure; rewrite on Django or another stack	Keep, subject to the two-day spike in Phase 0	Week 2 of Phase 0
Accounting depth	Sales accounting only; full ledger	Sales accounting with export to Zoho Books and Tally	Before Phase 2
Pricing metric	Per organisation; per user; hybrid	Per organisation with users included	Before Phase 3
Hosting	AWS Mumbai or Hyderabad; an Indian cloud provider	Choose by written quote. Require an India region, managed PostgreSQL and automated backups.	Start of Phase 0
Telephony	Build click-to-call now; integrate later	Later, through a cloud-telephony provider such as Exotel	After Phase 3
Mobile	Progressive web app; native app	PWA first	Start of Phase 1
Support model	Who answers customers, and how fast	Name one owner and a response-time target	Before Phase 3

Sources

Checked on 20 October 2026. Prices change, so confirm on the vendor's page before quoting a customer.

Opened and read

- Bigin by Zoho: official India price clarification
- Aaxonix Zoho pricing calculator, citing Zoho's March 2026 India price sheet
- Patron Accounting: Zoho Books pricing in India, 2026
- OEC.sh: Odoo pricing in India
- itforsme: Freshsales pricing in India
- itforsme: LeadSquared pricing in India

From search results only, not opened (treat as approximate)

- itforsme: Zoho People pricing in India
- Bar & Bench on the DPDP Rules 2025
- Uniqus on DPDP timelines
- Hogan Lovells on the DPDP Act and data processors